

HE2003 Econometrics I

Tutorial 2 (Week 3) – Problems & Solutions

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Overview

- **Lecture coverage:** Lecture 2 — *Linear Regression with One Regressor 1*: the simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$; the OLS minimisation problem; the first-order-condition (FOC) derivation of the closed-form OLS estimators; the summation identities that convert the “ $1/n$ ” form of $\hat{\beta}_1$ into the deviation-from-mean form; and the OLS predicted values and residuals.
- **Questions:** 5 questions from Tutorial Question Set No. 2.
- **Notation:** as in lectures — true unknown parameters β_0, β_1 ; generic candidate values b_0, b_1 (the arguments of the minimisation); OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ (random variables, functions of the sample); true error u_i ; OLS residual $\hat{u}_i$ (never e_i); ESS explained, SSR residual, $TSS = ESS + SSR$.

Useful rules (summation notation, restated verbatim from the question sheet).

Let $x_1, \dots, x_n$ and $y_1, \dots, y_n$ be two sequences of numbers and let a be a constant. Then

$$\sum_{i=1}^n ax_i = a \sum_{i=1}^n x_i, \quad (\text{R1})$$

$$\sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i, \quad (\text{R2})$$

$$\sum_{i=1}^n a = n \cdot a. \quad (\text{R3})$$

These three rules are all that is needed to expand every square and cross-term appearing below.

Question 1 (The idea of OLS estimation and the minimisation problem)

Problem

Explain the idea of OLS estimation and the related minimisation problem. (*Tutorial Question Set No. 2, Q1.*)

Solution

Method 1: Least-squares objective

- Model $Y_i = \beta_0 + \beta_1 X_i + u_i$: β_0, β_1 unknown; only $(X_i, Y_i)_{i=1}^n$ observed.
- Candidate line $b_0 + b_1 X$ has prediction error $Y_i - (b_0 + b_1 X_i)$ at observation i .
- Squaring stops errors cancelling and penalises large misses: $S(b_0, b_1) := \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$.
- OLS picks the (b_0, b_1) minimising S : the line closest to the data in squared vertical distance.

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2.$$

Question 2 (Notation: (β_0, β_1) vs. (b_0, b_1) vs. $(\hat{\beta}_0, \hat{\beta}_1)$)

Problem

Explain the difference of the notations (β_0, β_1) , (b_0, b_1) , and $(\hat{\beta}_0, \hat{\beta}_1)$ used in class. (*Tutorial Question Set No. 2, Q2.*)

Solution

Method 1: Role of each symbol

- (β_0, β_1) : true population intercept and slope; fixed, unknown constants.
- (b_0, b_1) : arbitrary candidate values; the arguments of $S(b_0, b_1) = \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$.
- $(\hat{\beta}_0, \hat{\beta}_1)$: the minimisers of S ; functions of the sample, hence random variables (value on one sample = estimate).

Question 3 (Deriving the OLS estimators from the FOCs)

Problem

Derive the formula of the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ following the steps in the lecture note. (*Tutorial Question Set No. 2, Q3.*)

Solution

Method 1: First-order conditions

$$S(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 \rightarrow \min_{b_0, b_1}$$

$$\frac{\partial S}{\partial b_0} = -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) = 0 \quad (\text{FOC } b_0)$$

$$\frac{\partial S}{\partial b_1} = -2 \sum_{i=1}^n (Y_i - b_0 - b_1 X_i) X_i = 0 \quad (\text{FOC } b_1)$$

Dividing by $-2n$ at $(\hat{\beta}_0, \hat{\beta}_1)$:

$$\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0 \quad (4)$$

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - \hat{\beta}_0 \bar{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0 \quad (5)$$

$$\implies \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad (4)$$

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} + \hat{\beta}_1 \bar{X}^2 - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0 \quad (\text{sub. } \hat{\beta}_0 \text{ in (5)})$$

$$\implies \hat{\beta}_1 = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2} \quad (X_i \text{ not all equal})$$

$$= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad (\text{Q4})$$

Second-order condition:

$$H = 2 \begin{pmatrix} \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \\ \sum_{i=1}^n X_i^2 & \sum_{i=1}^n X_i^3 \end{pmatrix}, \quad 2n > 0$$

$$\det H = 4 \left[n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2 \right] = 4n \sum_{i=1}^n (X_i - \bar{X})^2 > 0 \quad (\text{Q4})$$

$\implies H$ positive definite (X_i not all equal) $\implies$ unique global minimum

With $s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$, $s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$:

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X},}$$

Method 2: Centred regressor

$Y_i = \alpha + \beta_1(X_i - \bar{X}) + u_i$, $\alpha = \beta_0 + \beta_1\bar{X}$; minimise $T(a, b_1) = \sum_{i=1}^n [Y_i - a - b_1(X_i - \bar{X})]^2$.

$$\frac{\partial T}{\partial a} = -2 \sum_{i=1}^n [Y_i - \hat{a} - \hat{b}_1(X_i - \bar{X})] = 0$$

$$\Rightarrow \sum_{i=1}^n Y_i - n\hat{a} = 0 \quad \left(\sum_{i=1}^n (X_i - \bar{X}) = 0 \right)$$

$$\Rightarrow \boxed{\hat{a} = \bar{Y}}$$

$$\frac{\partial T}{\partial b_1} = -2 \sum_{i=1}^n [Y_i - \bar{Y} - \hat{b}_1(X_i - \bar{X})](X_i - \bar{X}) = 0$$

$$\Rightarrow \hat{b}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \hat{\beta}_1$$

$$\hat{\beta}_0 = \hat{a} - \hat{\beta}_1\bar{X} = \bar{Y} - \hat{\beta}_1\bar{X} \quad (\beta_0 = \alpha - \beta_1\bar{X})$$

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}.}$$

Question 4 (Two summation identities)

Problem

Show the following holds:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}, \quad \text{where } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

(Hint: show $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$ and $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$.) (Tutorial Question Set No. 2, Q4.)

Solution

Method 1: Direct expansion

Part (a).

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + \sum_{i=1}^n \bar{X}^2 && \text{(R1, R2)} \\ &= \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 && \left(\sum_{i=1}^n X_i = n\bar{X}; \text{R3}\right) \\ &= \boxed{\sum_{i=1}^n X_i^2 - n\bar{X}^2}. \end{aligned}$$

Part (b).

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^n X_i Y_i - \bar{Y} \sum_{i=1}^n X_i - \bar{X} \sum_{i=1}^n Y_i + \sum_{i=1}^n \bar{X}\bar{Y} && \text{(R1, R2)} \\ &= \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y} - n\bar{X}\bar{Y} + n\bar{X}\bar{Y} && \text{(R3)} \\ &= \boxed{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}. \end{aligned}$$

Ratio.

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^n X_i^2 - n\bar{X}^2} \quad \text{(a), (b), then } \div n \text{ top and bottom}$$

$$\boxed{\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}}.$$

Method 2: Using $\sum_{i=1}^n (X_i - \bar{X}) = 0$

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^n (X_i - \bar{X})Y_i - \bar{Y} \sum_{i=1}^n (X_i - \bar{X}) && \text{(R1, R2)} \\ &= \sum_{i=1}^n X_i Y_i - \bar{X} \sum_{i=1}^n Y_i && \left(\sum_{i=1}^n (X_i - \bar{X}) = 0 \right) \\ &= \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y} && \text{(b)}\end{aligned}$$

Part (a): put $Y_i = X_i$ ($\bar{Y} = \bar{X}$).

Question 5 (Class size and test scores: numerical predictions)

Problem

Suppose that a researcher, using data on class size (CS) and average test scores from 50 third-grade classes, estimates the OLS regression

$$\widehat{\text{TestScore}} = 640.3 - 4.93 \times \text{CS}.$$

- A classroom has 25 students. What is the OLS regression's prediction for that classroom's average test score?
- Last year a classroom had 21 students, and this year it has 24 students. What is the OLS regression's prediction for the change in the classroom average test score?
- The sample average class size across the 50 classrooms is 22.8. What is the sample average of the test scores across the 50 classrooms? (*Hint: review the formulas of the OLS estimators.*)

(Tutorial Question Set No. 2, Q5.)

Solution

Part (a): prediction at CS = 25

$$\widehat{\text{TestScore}} = 640.3 - 4.93 \times 25 \implies \boxed{\widehat{\text{TestScore}} = 640.3 - 123.25 = 517.05.}$$

Part (b): predicted change from CS = 21 to CS = 24

Method 1: predict both years, then subtract

$$\widehat{\text{TestScore}}_{\text{last year}} = 640.3 - 4.93 \times 21 = 536.77$$

$$\widehat{\text{TestScore}}_{\text{this year}} = 640.3 - 4.93 \times 24 = 521.98$$

$$\Delta \widehat{\text{TestScore}} = 521.98 - 536.77 \implies \boxed{\Delta \widehat{\text{TestScore}} = -14.79 \text{ points.}}$$

Method 2: slope as marginal effect

$$\Delta \widehat{\text{TestScore}} = [\hat{\beta}_0 + \hat{\beta}_1 \text{CS}_2] - [\hat{\beta}_0 + \hat{\beta}_1 \text{CS}_1] = \hat{\beta}_1 \Delta \text{CS}$$

$$\boxed{\Delta \widehat{\text{TestScore}} = \hat{\beta}_1 \Delta \text{CS} = -4.93 \times 3 = -14.79 \text{ points.}}$$

Part (c): sample average test score, given mean CS = 22.8

$$\begin{aligned}\overline{\text{TestScore}} &= \hat{\beta}_0 + \hat{\beta}_1 \overline{\text{CS}} & (\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \text{ Q3}) \\ &= 640.3 - 4.93 \times 22.8 = 640.3 - 112.404\end{aligned}$$

$\overline{\text{TestScore}} = 527.896 \text{ points.}$

Final Answers

1. $(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$.
2. β : true unknown parameters; b : candidate arguments; $\hat{\beta}$: minimisers, random variables (estimates on one sample).
3. $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, $\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$; Hessian positive definite.
4. $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$; $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$.
5. (a) 517.05; (b) -14.79 points; (c) 527.896 points.